

Influence of hydrophobicity and roughness on the wetting and flow resistance of water droplets on solid surface: A many-body dissipative particle dynamics study

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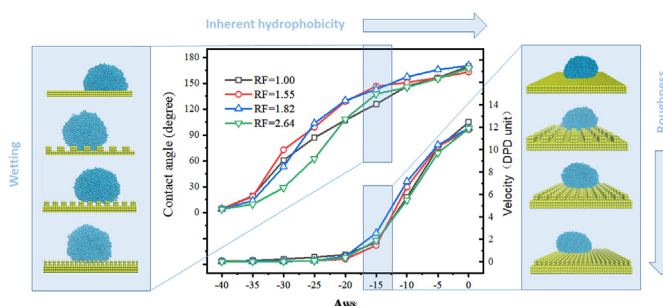
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HIGHLIGHTS

- The wetting and flow behaviors of water droplet on smooth and rough solid surfaces are simulated by the MDPD method.
- The influence of inherent hydrophobicity and roughness on apparent hydrophobicity of surface are studied systematically.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrophobicity and roughness are crucial properties for determining rock wetting and flow resistance. The wetting and flow behavior of water droplets on smooth and rough solid surfaces are investigated by performing a many-body dissipative particle dynamics simulation. The results show that, in terms of the roughness, a solid surface can be divided into three regions. In Region I, as inherent hydrophobicity increases, roughness can enhance the apparent hydrophilicity of a hydrophilic surface. In Region II, the increased hydrophobicity improves the apparent hydrophobicity of a more hydrophobic surface. In Region III, when the hydrophobicity is beyond the superhydrophobic boundary, the enhancement effect of the rough surface on the apparent hydrophobicity is weakened or even disappears. With an increase in the roughness, the boundaries among these three regions and their ranges show nonmonotonic changes, making the contact mode (Wenzel state/Cassie and Baxter state), contact angle hysteresis and drag reduction effect change nonmonotonically.

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1. Introduction

The wetting phenomena on rough surfaces are ubiquitous in nature and industries. Different fluids and rocks exhibit different wetting. Hydrophobicity and roughness are crucial properties that determine rock wetting and flow resistance. They are fundamental

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properties of a surface, and play an important role in many biological and industrial processes (Xu et al., 2019; Wang and Gong, 2017; Xu et al., 2019; Feng and Jiang, 2006; Wang et al., 2019; Tian et al., 2013). In oil and gas field development, rock wettability has an important impact on the fluid flow process. In the low-permeability oilfields, injecting working fluid into the formation is difficult because the injection pressure is much higher during water flooding. This phenomenon results in a severe shortage of energy supplements and leads to oil production far below expectations. An alternative method to enhance oil recovery is to inject nanoparticles into the formation, in which nanoparticles are dispersed into the porous media, forming an adsorption layer on the rock surface. Scanning electron microscopy (SEM) revealed that the adsorption of the nanoparticles form a rough secondary structure similar to that of lotus leaves (Gu et al., 2007). This type of rough surface is usually strongly hydrophobic or even superhydrophobic, and therefore effectively reduces the water injection resistance (Hendraningrat and Torsæter, 2014). The rough surface was proven to affect the structural disjoining pressure, which would accelerate the spreading of the droplets, causing the oil droplets to gradually detach from the solid surface (Wasan and Nikolov, 2003; Wasan et al., 2011; Liu et al., 2012). In addition, the surface roughness could considerably reduce the friction between the fluid and the boundaries, and the slippage of the fluid at the channel boundary is greatly increased at the same time (Cottin-Bizonne et al., 2003; Rodriguez et al., 2009).

Young's angle, which is defined as the contact angle of a droplet contacting an ideal flat surface, is adopted to characterize the inherent hydrophobicity of a surface (Young, 1805). For a real surface, it cannot be ideally smooth and it might become somewhat rough because of scraping and corroding (Guo et al., 2008). Thus, the apparent contact angle is usually different from Young's angle owing to the roughness (Jeong et al., 2012).

It is well-known that roughness can make hydrophilic surfaces more hydrophilic while making hydrophobic surfaces more hydrophobic (Quéré, 2008; Rothstein, 2009). Because of the different roughness and surface chemistry, water might fill up or stay on top of the space between the structures on the surface, as illustrated by the theories proposed by Wenzel (Wenzel, 1936) and Cassie and Baxter (Cassie and Baxter, 1944) (see Fig. 1A). In some cases, a portion of the water falls between the textured structures of a rough solid surface, which is neither the Cassie-Baxter state nor the Wenzel state but a mixed wetting state (Marmur, 2003).

The roughness of a solid surface can be characterized by the normalized surface roughness factor (RF) and root-mean-square roughness (R) (Zhou et al., 2019). RF is defined as the ratio of the actual solid surface area to the projected solid surface area (Jeong et al., 2012; Li et al., 2014). Different materials usually have diverse surface chemical properties and roughness, making it difficult to quantify and compare the effect of roughness on the apparent hydrophobicity of different materials. The contact angle of water on the flat surface of most materials is typically lower than 180° (Lafuma and Quéré, 2003); while some rough surfaces can become very close to 180° (Öner and McCarthy, 2000).

Studies have focused on dynamic hydrophobicity instead of static hydrophobicity to improve the understanding of the specific behavior of water droplet detachment from a microstructured superhydrophobic surface (Yao et al., 2020; Lv et al., 2010; Miwa et al., 2000; Kong et al., 2019). A rationally designed surface that considers the shape and extent of the three-phase line has been proven to be a powerful way to facilitate a functional superhydrophobic surface that enhances the sliding behavior of water droplets over pillar-like and groove structures (Yoshimitsu et al., 2002). In addition to experimental approaches (Öner and McCarthy, 2000; Yao et al., 2020; Lv et al., 2010; Miwa et al., 2000; Kong et al., 2019; Yoshimitsu et al., 2002; Wang et al.,

2017; Olin et al., 2013; Schellenberger et al., 2016; Huhtamäki et al., 2018); molecular simulations have been used in many studies to investigate the wetting properties. These studies have revealed the influence of the solid surface structure on the apparent wettability at the atomic scale (Ou et al., 2019; Yoo et al., 2018; Yaghoubi and Foroutan, 2018; Kwon et al., 2018; Ambrosia and Ha, 2018; Liu et al., 2018; Shi and Dhir, 2009). Meanwhile, many studies have also investigated the dynamic characteristics of moving water droplets on different surfaces, and analyzed the drop shape change and the effect of applied force (Jeong et al., 2012; Li et al., 2014; Li et al., 2018; Liu et al., 2018; Ritos et al., 2013). However, the simulation models used in previous studies either did not cover the hydrophobicity ranging from 0 to 180° or did not analyze the surface roughness in detail via RF or R parameterization. Thus, a comprehensive understanding of the relationship between roughness and hydrophobicity is lacking. Moreover, the effect of roughness and hydrophobicity on the flow resistance has rarely been discussed in these studies.

In this study, the wetting and flow behaviors of droplets on solid surfaces with different roughness and hydrophobicity were investigated using the many-body dissipative particle dynamics (MDPD) method. The main purpose of this study was to understand the influence of different roughness values on the apparent hydrophobicity, wetting state, and flow resistance of a solid surface. To achieve this goal, we first simulated the contact angles of different hydrophobic surfaces on smooth surfaces in a static state, establishing the relationship between the interaction parameters of the MDPD model and the solid surface hydrophobicity. Next, the static contact angle, dynamic contact angle, and flow velocity of the droplet on a solid surface with the same R but different RF were simulated. Furthermore, the influence of RF on the apparent contact angle, wetting mode, contact angle hysteresis, and flow resistance of a droplet on a solid surface with different inherent hydrophobicities was evaluated.

2. Methods

2.1. MDPD method

MDPD, a many-body modification method of dissipative particle dynamics (DPD) (Chen et al., 2010); has been developed to simulate the vapor–liquid interface and fluid in capillaries (Zhou et al., 2019; Chen et al., 2010; Chen et al., 2012; Chen et al., 2014). Within the MDPD method, forces ($\mathbf{F}_{ij}$) between beads include the conservative ($\mathbf{F}_{ij}^C$), dissipative ($\mathbf{F}_{ij}^D$), and random ($\mathbf{F}_{ij}^R$) parts as shown in Eq. (1) below.

$$\mathbf{F}_{ij} = \mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R \quad (1)$$

In contrast to the conventional DPD method originally proposed by Hoogerbrugge and Koelman (Hoogerbrugge and Koelman, 1992); the conservative ($\mathbf{F}_{ij}^C$) includes the attractive force ($A_{ij} < 0$) and the repulsive force ($B > 0$), as shown in Eq. (2).

$$\mathbf{F}_{ij}^C = A_{ij}w_C(r_{ij})\mathbf{e}_{ij} + B(\bar{\rho}_i + \bar{\rho}_j)w_d(r_{ij})\mathbf{e}_{ij} \quad (2)$$

where ρ_i and ρ_j are the local density, and $w_C(r_{ij})$ is a weight function that vanishes for $r > r_d = 0.75$. This modification makes it possible to simulate the wetting behavior in the vapor–liquid interface by adjusting the interaction parameter A_{ij} .

2.2. MDPD models

As for the MDPD model, one water bead (W) consisted of three water molecules (Fig. 2A). Three water drops of different sizes contained 1000, 2000, and 4000 W beads. The solid surface was con-

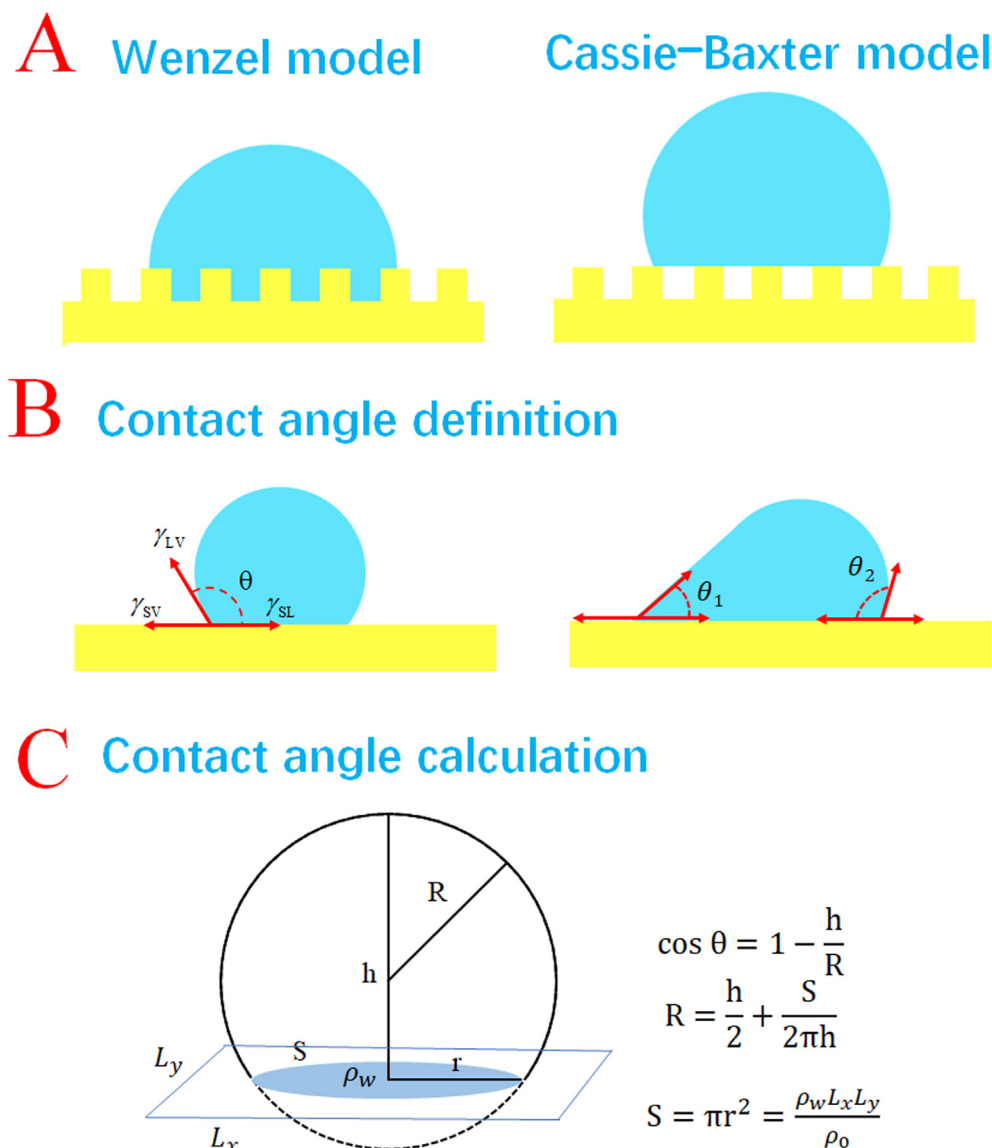


Fig. 1. Wetting and contact angle measurement. (A) Wenzel model and Cassie-Baxter model; (B) Definition of contact angle. (C) Calculation method of contact angle (θ). The height (h) and the radius (r) of droplet contact surface can be obtained from the density profile, where S is the contact surface area, ρ_w is the water density value on the contact surface in vertical direction, ρ_0 is the density value of the bulk water, and L_x and L_y are the length of the sides of the solid surface.

structed using the S bead. The smooth solid surface had a size of $48 \times 48 \times 1.5$ DPD length units. As for the rough surface, the height of the protrusion plate was 1.80 DPD length units. The width of the protrusion plate is equal to the cavity between them, which makes the root-mean-square roughness of all the rough surfaces identical (0.90). In contrast, the root-mean-square roughness of the smooth solid surface is 0.00. When the roughness of the solid surface is represented by the roughness factor (RF), the RF value for the smooth solid surface is 1.00, while RF values for the rough surfaces are 1.55, 1.82, and 2.64.

2.3. Simulation method

Equilibrium many-body dissipative particle dynamics (EMDPD) simulations were performed using the DL_MESO software package (Seaton et al., 2013). In all simulations, the cutoff $r_c = 1$ and $r_d = 0.75$, and the time step was set to be 0.01 DPD time units. The water droplets were initially suspended above the center of the solid surface. A gravity force of 0.02 DPD force units was applied to all the beads in the Z direction. In the nonequilibrium many-body dissipative particle dynamics (NEMDPD) simulations,

an external driving force ($Fe = 0.02$ DPD force units) was applied on the W beads in the horizontal direction to push the water droplet. Therefore, the model can also be regarded as a sloping surface with an angle of 45° , in which the droplet is subjected to 1.44 times the gravity.

2.4. Wetting angle measurement and velocity analysis

The contact angle can be obtained in two ways: (1) direct measurement according to the definition (see Fig. 1B) and (2) calculation based on the density distribution (see Fig. 1C). The cross-sectional area between the water and solid surface can be calculated from the water density profile. The density of water was taken as $\rho_0 = 6.05$, while the density of water at the cross-section is ρ_w . The total solid surface area was $L_x L_y$, and the cross-sectional area was $S = \rho_w L_x L_y / \rho_0$. The radius of the water droplet (R) and the contact angles can be calculated based on the equations shown in Fig. 1C.

The contact angles of water droplets of different sizes were simulated and compared (see Fig. S1). The results showed that the difference in the contact angle caused by different droplet sizes was

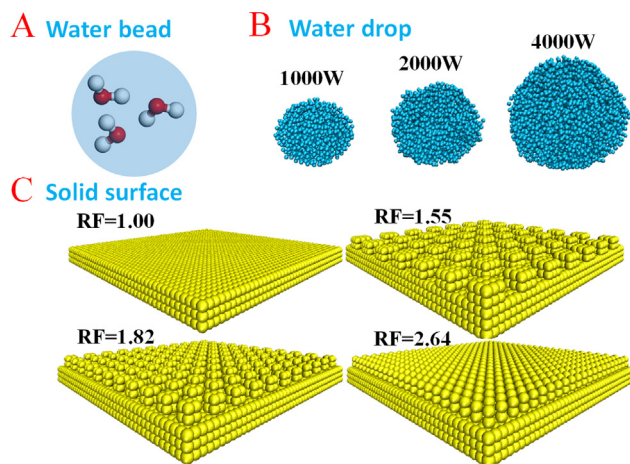


Fig. 2. MDPD models of water and solid surface. (A) MDPD model of water (W bead) containing 3 water molecules; (B) MDPD model of water droplet with 1000, 2000 and 4000 W beads; (C) Solid surfaces with different RF. The RF values were set as 1.00, 1.55, 1.82, and 2.64. The height of the protrusion plate is set as 1.80. The width of the protrusion plate is equal to the cavity between them, which makes the root-mean-square roughness of all the rough surfaces identical.

less than that of A_{WS} (interaction parameter between W bead and S bead) values. This indicates that the effect of the droplet size can be ignored. The contact angle was obtained by measuring the 4,000 W water droplets to reduce the error when measuring the advancing and receding contact angles. When the solid surface was hydrophilic, and the contact angle was very small, i.e., close to 0° when water droplets spread on the solid surface, the contact angle was measured from the 1,000 W water droplets. The density distributions of the static and moving droplets at different time points along the vertical direction are shown in Figs. S2 and S3. There was a short period (less than 100 DPD time units) of structural

vibration at the beginning. Subsequently, the density distribution tended to balance. In the NEMDPD simulations, there was an accelerated motion of the droplet at the beginning. After 400 DPD time units, the speed increment became smaller and closer to uniform motion (see Fig. S4). Therefore, only the data after 400 DPD time units were collected and used in the contact angle and velocity calculations.

3. Results

3.1. Wetting angles on smooth surface

To correlate the interaction parameter in the MDPD model with the surface chemistry, we first simulated the wetting behavior of a water droplet on a smooth solid surface. A_{WS} ranges from -40 to 0 to represent different inherent hydrophobicities. When $A_{WS} = A_{WW} = -40$, the solid surface had the strongest affinity for water. When $A_{WS} = 0$, the solid surface had no attraction interaction with water, exhibiting absolute hydrophobicity.

As shown in Fig. 3, the contact angle changed from $\sim 0^\circ$ to $\sim 180^\circ$ as the A_{WS} ranged from -40 to 0 . The equilibrium contact angle on the solid surface with different hydrophobicity were calculated based on the density profile of water (see Fig. 4A). The contact angle and the A_{WS} values had an almost linear relationship. These results indicate that a solid surface with different inherent hydrophobicity could be effectively simulated by adjusting the attraction force between the water and solid surface in the MDPD model.

3.2. Wetting angles on rough surface

Using the MDPD model, the contact angles of water droplets on rough surfaces were simulated, and the density profiles were ana-

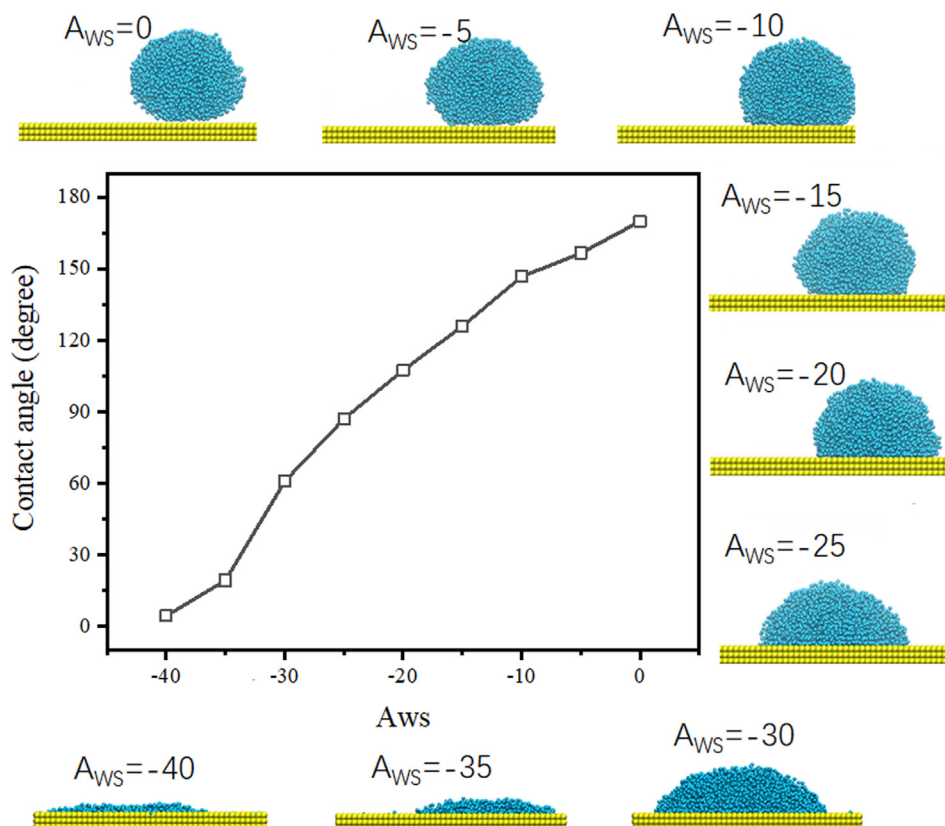


Fig. 3. Wetting angles of water droplets on the smooth surface.

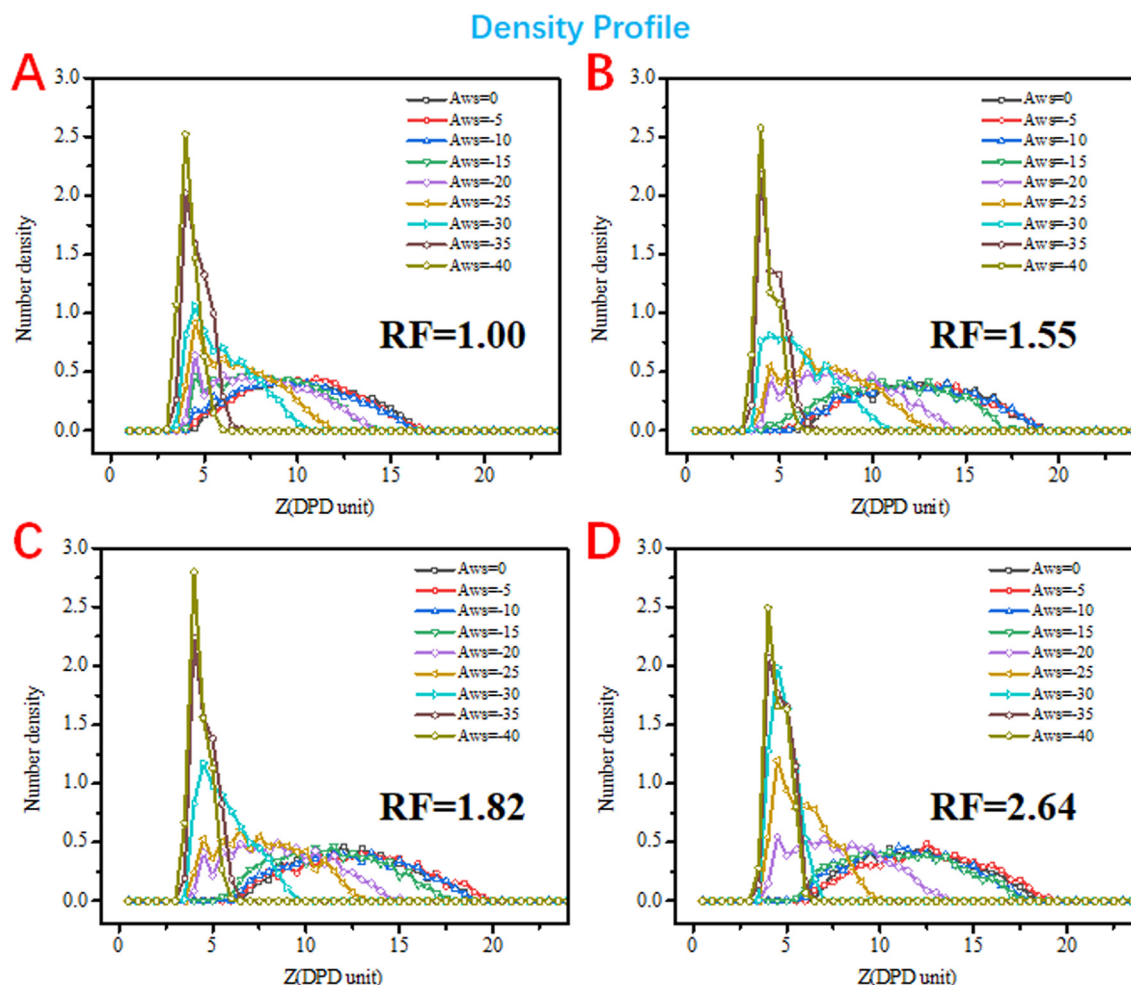


Fig. 4. Density profiles of the static water droplets on the smooth and rough solid surfaces with different hydrophobicity. (A) Smooth solid surface ($RF = 1.00$) and rough solid surfaces with RF values of (B) 1.55, (C) 1.82, and (D) 2.62.

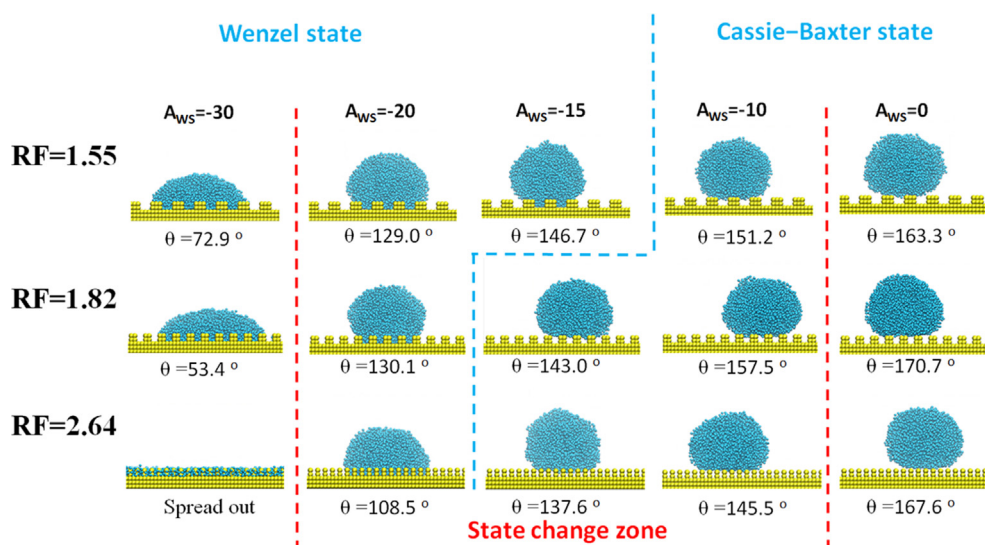


Fig. 5. Snapshots of the static water droplets on the rough solid surface with different hydrophobicity.

lyzed. Compared with the smooth solid surface, the distribution peak of water on the rough hydrophilic surface was higher and sharper, while the distribution peak of water on the rough hydrophobic surface was lower and wider (see Fig. 4B, 4C, 4D

and S5-S7). As shown in Fig. 5, with increased hydrophobicity, the contacting mode of water droplets changed from the Wenzel state to the Cassie and Baxter state. It should be noted that with an increase in RF values, the transition region between these two

states changed. When $RF = 1.55$, a transition occurred between $A_{WS} = -15$ and $A_{WS} = -10$. When the RF was greater than 1.82, the transition occurred between $A_{WS} = -20$ and $A_{WS} = -15$. When the RF increased from 1.55 to 1.82, the transition region between these two states moved to a more hydrophilic region (corresponding to the intrinsic hydrophobicity accounted for by A_{WS}). Although the transition region accounted for by A_{WS} remained the same, the apparent contact angle became smaller (corresponding to the apparent hydrophobicity) as RF increased from 1.82 to 2.64.

3.3. Wetting angles of moving droplet

The dynamic hydrophobicity was then investigated by observing the advancing contact angle and receding contact angle of the water droplets in the flow state. As shown in Figs. 6, 7 and S8–S11, in the hydrophilic region, the advancing contact angle was larger than the receding contact angle. This phenomenon is named “Contact angle hysteresis”, which is caused by the slower velocity of the advancing part of the water droplet than that of

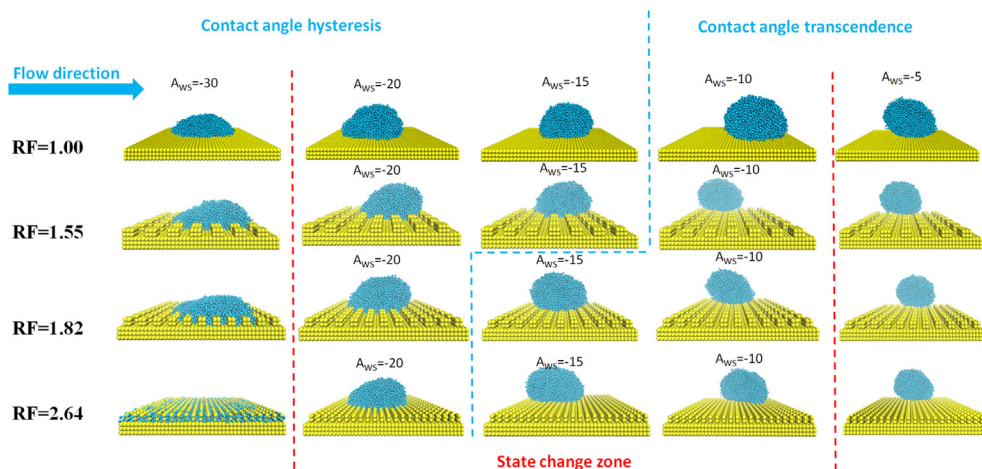


Fig. 6. Snapshots of the moving water droplets on the smooth and rough solid surfaces with different hydrophobicity.

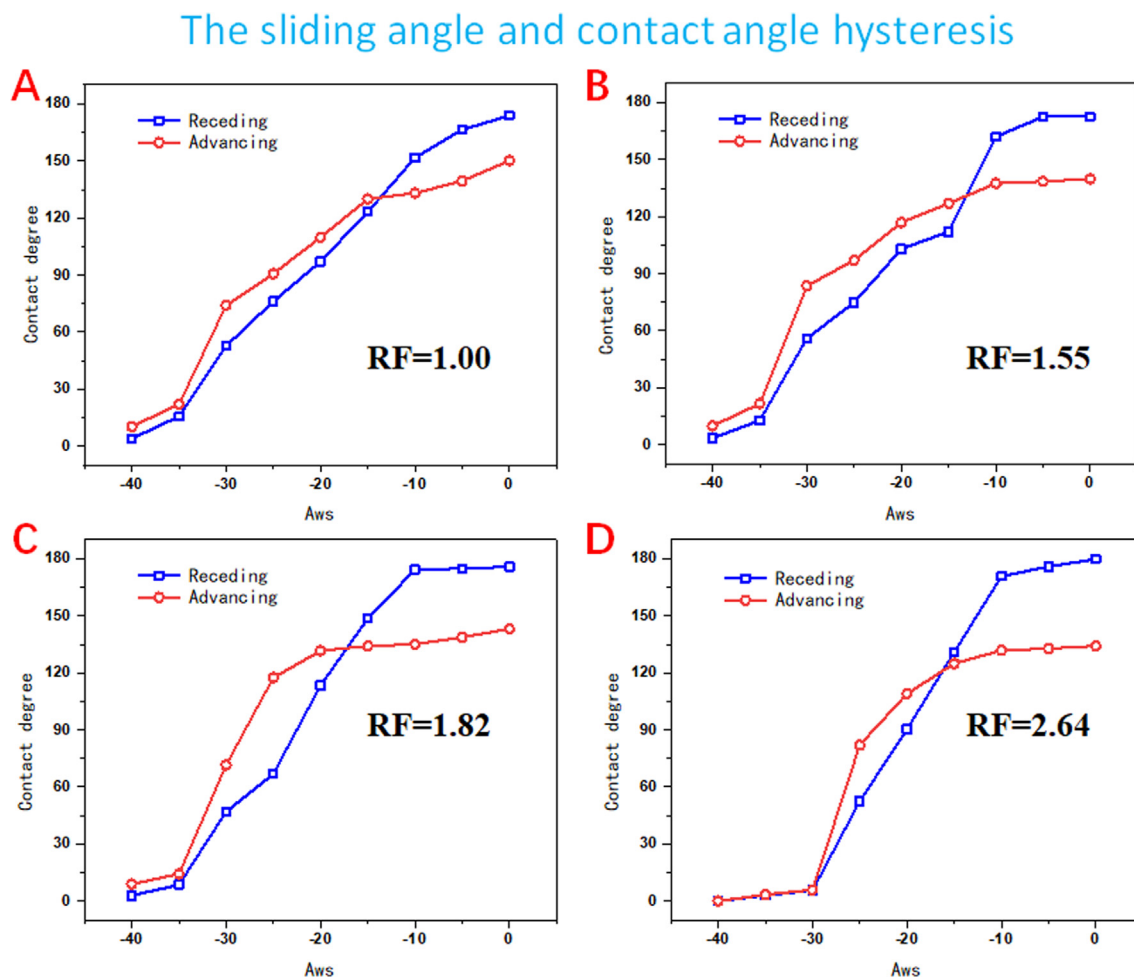


Fig. 7. Advancing and receding contact angles of moving water droplet on the smooth and rough solid surfaces with different hydrophobicity.

the receding part (Yeh et al., 2008). When the solid surface became increasingly hydrophobic (between $A_{WS} = -15$ and $A_{WS} = -10$ for the smooth solid surface), the advancing contact angle became smaller than the receding contact angle, indicating that the part of the droplet near the solid surface had a higher velocity. This contact angle transcendence phenomenon was caused by the velocity difference along the vertical direction. To the best of our knowledge, this transcendence phenomenon has not yet been reported in experimental studies. It is possibly because that in most cases, even if the solid surface is superhydrophobic, an adhesion force still exists between the droplet in contact with the solid surface, which would hinder the movement of the droplet. The simulation results predicted that if the velocity was high, the temperature was not low, and the adhesion force between the droplet and the solid surface was sufficiently reduced, the friction within the liquid molecules caused by thermal dynamics would become larger than the friction from the liquid–solid interface. To ensure this, we performed two additional simulation runs. As shown in Fig. S12, when the temperature was lower (i.e., $T = 276$ K), the advancing contact angle was almost equal to the receding contact angle. When the external force (F_e) is lower, i.e. $F_e = 0.001$ DPD force units, the advancing contact angle became larger than the receding contact angle at a low velocity. Compared with the smooth solid surface, this transition between contact angle hysteresis and transcendence occurred at a more hydrophilic region (between $A_{WS} = -20$ and $A_{WS} = -15$) when the RF of the rough surface was greater than 1.82.

4. Discussion

4.1. Effect of roughness on wettability

Previous studies have shown that rough surfaces could make hydrophobic surfaces more hydrophobic and hydrophilic surfaces more hydrophilic. MDPD simulation results showed that roughness could not always enhance the apparent hydrophilicity of a hydrophilic surface or the apparent hydrophobicity of a hydrophobic surface, depending on both the roughness value and the inherent surface hydrophobicity. As shown in Fig. 8B, when $RF = 1.55$, the effect of decreasing the contact angle on the hydrophilic region ($A_{WS} < -35$) was not significant, while increasing the contact angle in a more hydrophobic region ($-30 < A_{WS} < -10$) was significant. It should be noted that when $RF = 1.55$ and $A_{WS} = -30$, the surface roughness increased the contact angle of the hydrophilic surface (from 60.9° to 72.9°), possibly because of the limitation on the spreading of the droplet along the entire surface caused by the hydrophilic side of the protrusion plate. When $RF = 1.82$, the contact angle decreased significantly in the hydrophilic region ($A_{WS} < -30$), and the increase of contact angle in the hydrophobic region ($-25 < A_{WS} < -5$) is higher than that in the case of $RF = 1.55$. When $RF = 2.64$, the decrease in contact angle in the hydrophilic region ($A_{WS} < -25$) was larger than that in $RF = 1.82$, while the effect of increasing the contact angle in the hydrophobic region ($-20 < A_{WS} < -15$) was not significant, which is less than that of $RF = 1.55$ and RF

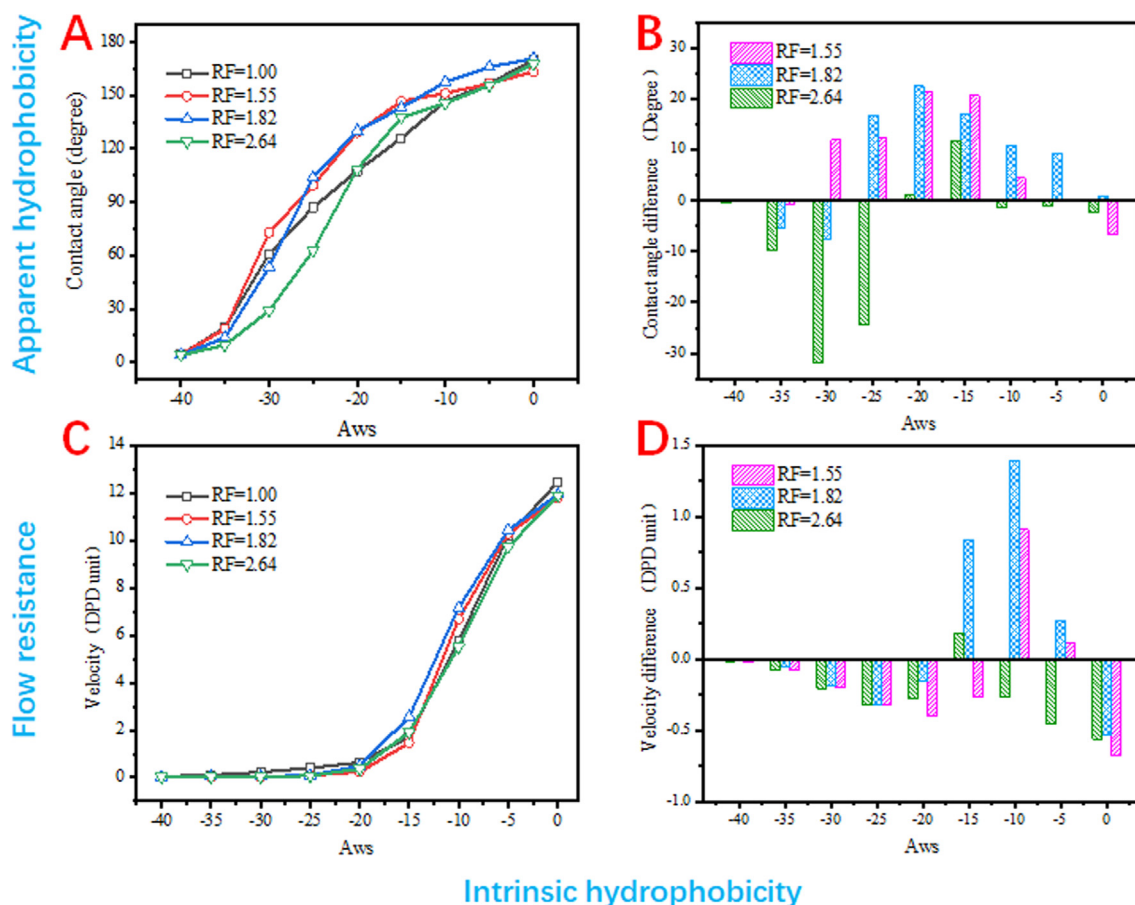


Fig. 8. Apparent contact angle of static water droplet and flow velocity of moving water drop. (A) Contact angle profile of static water drop. (B) Contact angle difference of static water droplet between smooth and rough solid surface. (C) Velocity profile of moving water drop. (D) Velocity difference of moving water droplet between smooth and rough solid surface.

= 1.82. It should be noted that when $A_{WS} = 0$, almost all of the rough solid surfaces lose the ability to increase the contact angle further. A possibly reason for this is that the hydrophobicity of a smooth solid surface had reached a rather high level that was hard to overcome.

4.2. Effect of roughness on flow resistance

The variation in the velocity of the flowing droplet on a solid surface with different inherent hydrophobicity was similar to that of the apparent contact angle of the droplet in the static state. Generally, with an increased roughness, the apparent contact angle decreased and the flow resistance increased (the velocity decreased) in the hydrophilic region, whereas in the hydrophobic region, the contact angle increased and the flow resistance decreased (the velocity increased).

Compared with the smooth surface, the influence of the rough surface on the flow velocity in most cases increased the flow resistance, rather than presenting a drag reduction effect: when $RF = 1.55$, drag reduction occurred in the superhydrophobic region ($-10 < A_{WS} < -5$). When $RF = 1.82$, the drag reduction region became wider ($-15 < A_{WS} < -5$). When $RF = 2.64$, the drag reduction region became very narrow and occurred only when $A_{WS} = -15$.

The transition boundaries of the three drag reduction regions affected by roughness moved toward more hydrophobic regions, unlike the apparent contact angle in the static state. A possible reason is that the flow resistance of a droplet on a solid surface was not only affected by the apparent hydrophobicity, but also affected by the friction of the rough surface. The results showed that surfaces with the most significant increase in static contact angle were not with the most obvious drag reduction effect; the region with the most significant drag reduction effect was more hydrophobic than the region with the largest increase in apparent contact angle in the static state.

In the two extreme regions, such as $A_{WS} = A_{WW} = -40$, the flow resistance of the smooth and rough surfaces reached a high level. The water stopped flowing and was completely locked by the surface, making the difference in velocity between smooth and rough surfaces small. As A_{WS} approached 0, the apparent contact angle of the rough surface did not increase further compared with that of the smooth surface. Conversely, an increase in roughness enhanced the friction between the droplet and solid surface, resulting in larger flow resistance.

5. Conclusion

This study investigated the wetting and flow behaviors of water droplets on smooth and rough solid surfaces using the MDPD method. The MDPD models were built with identical root-mean-square roughness (R) but different normalized surface roughness factors (RF) to investigate the effect of roughness on apparent hydrophobicity. The parameters of the MDPD model were correlated with the inherent hydrophobicity by establishing the mapping relationship between the interaction parameters (A_{WS}) and static contact angle ranging from $\sim 0^\circ$ to $\sim 180^\circ$. The results indicated that with an increased inherent hydrophobicity, the increased roughness did not always enhance the apparent hydrophilicity of a hydrophilic surface or enhanced the apparent hydrophobicity of a hydrophobic surface. When the hydrophobicity was beyond the superhydrophobic boundary, the enhancement effect of the rough surface on the apparent hydrophobicity was weakened or even disappeared. On the rough surface, the contact mode between the water droplet and the surface changed from the Wenzel state to the Cassie and Baxter state with an increased inherent hydrophobicity. In addition, the critical inherent

hydrophobicity boundary of the wetting transition varied with different roughness values. In the superhydrophobic region, the contact angle hysteresis was weakened and even reversed: the advancing angle became smaller than the receding angle. Furthermore, the influence of the rough surface on the flow velocity in most cases increased the flow resistance. Compared with the roughness-caused enhancement region of the apparent contact angle in static state, the drag reduction effect only occurred in a narrow region in dynamic state. Furthermore, the optimal drag reduction regions moved toward more hydrophobic regions than the optimal apparent contact angle enhancement regions, possibly due to the friction effect. In other words, the apparent static hydrophobicity and dynamic hydrophobicity were highly dependent on both the inherent hydrophobicity and surface roughness, representing a nonmonotonic change with increased roughness.

CRediT authorship contribution statement

Qingjun Du: Writing – original draft, Methodology. **Peng Zhou:** Supervision, Conceptualization, Writing – review & editing. **Yuping Pan:** Visualization, Investigation. **Xiao Qu:** Validation. **Lu Liu:** Data curation. **Hui Yu:** Software. **Jian Hou:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2021.117327>.

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